

Growing up in a village in Bihar where education was often considered a luxury, I learnt early that knowledge transforms lives. This belief drove me to earn a bachelor's in computer science (honours) from Panjab University, with department rank 1. As I delved deeper into AI through Stanford courses and research projects, I became captivated by a fundamental challenge: real-world tasks rarely fit into a single modality. Whether summarising content from news articles with accompanying images or answering questions from educational lectures, meaningful reasoning requires integrating text, visuals, audio, and temporal cues. These experiences shaped my passion for multimodal AI and developing systems that can jointly reason over different media types – a vision I aim to pursue through advanced study at ETH Zurich.

Having worked across six labs in three countries, including CMU and other leading institutes, I have built a strong foundation in AI, resulting in 6 publications at top venues with over 70 citations. My research foundation began at IIT Patna under Prof. Sriparna Saha, where I developed a system to identify product complaints from customer reviews. Instead of analysing text alone, I proposed a multimodal Bitransformer framework that combined the review text with visual product data in a two-stage architecture. This study showed how complementary modalities strengthen understanding since accuracy improved from 81.34% to 84.57%. Our work was accepted at the European Conference on Information Retrieval (ECIR) 2023. Based on this insight, I extended my work on multimodal summarisation in Indian languages with Prof. Saha through the ACM IKDD UPLINK Research Internship Program. I adapted the BART model to incorporate a visually aware encoder that combined text and image data to achieve a ROUGE-1 score of 0.266 on Hindi datasets. Generating multi-output summaries was a challenge, and hence, I proposed an image pointer method that achieved 44.1% precision. This research was later published at the International Conference on Document Analysis and Recognition (ICDAR) 2024. Both these projects helped me to understand that to develop effective multimodal systems, careful architectural designs are required to capture cross-modal relationships.

At CMU's Xu Lab under Prof. Min Xu, I came across another dimension of multimodal learning: 3D cryo-ET classification using self-supervised learning. The difficulty in this was adapting Momentum Contrast for Unsupervised Visual Representation Learning (MoCo), originally designed for 2D images, to higher dimensionalities of 3D volumetric data in open-set environments presented. So, I proposed a 3D-tailored pipeline that integrated MoCo with Video Vision Transformer (ViViT) and Vision Transformer (ViT), improving performance and training stability. Working on 3D data without labels taught me how self-supervised methods could learn meaningful representations across modalities, with our classifier based on ViViT reaching an F1 score of 70.28% on real dataset classes. This work, which earned its spot in the International Conference on Computer Vision (ICCV) submission, furthered my understanding of representation learning across diverse data types. But the one research that most shaped my learning trajectory was at IIT Kharagpur under Prof. Somak Aditya, where I improved student learning by understanding context from long-form educational videos using large language models. Our 1,400 manually annotated examples from NPTEL lectures were insufficient, so I designed a question-generation pipeline using Google Gemini 1.5 Flash that took our dataset to almost 20,000 questions on varied reasonings and visual understanding scenarios. Following rigorous validation, I prepared a high-quality dataset of 4,300 question-answer pairs and spearheaded the benchmarking of several state-of-the-art multimodal LLMs, including VideoChatGPT and VideoLLaMA, in zero-shot and fine-tuned settings. This project, accepted at EMNLP 2025, crystallised my conviction that multimodal AI has the potential to democratise education but also revealed deficiencies in my knowledge related to scaling these systems and extending their reasoning capabilities.

Currently, I work as a Machine Learning Engineer at Recursive Softpro, designing production-ready multimodal RAG pipelines that fetch information from text, images, audio, and videos to create context-aware responses using GPT-4o. I have designed multi-agent orchestration systems and worked on an AI marketplace with custom workflows. While this experience has taught me to balance performance, latency, and reliability in the deployed systems, I realise that advancing the fundamental capabilities of multimodal reasoning, improving cross-modal attention mechanisms, developing better fusion architectures, and enabling more sophisticated visual-linguistic understanding require deeper theoretical grounding and sustained research focus that only a rigorous programme can provide.

ETH Zurich's MSc in Computer Science, specialising in machine intelligence, provides just that foundation. Courses such as Computer Vision, AI in Education, and Natural Language Processing will strengthen my theoretical understanding of the core modalities I work with. I am particularly interested in the research work of Prof. Mrinmaya Sachan, including "Can Vision-Language Models Solve Visual Math Equations?", since it directly aligns with my educational video QA research at IIT Kharagpur. Both research studies investigate how models reason over visual and textual information in educational contexts. Similarly, Prof. Michael Moor's work on "Reverse Image Retrieval Cues Parametric Memory in Multimodal LLMs" resonates with my RAG pipeline development, since both strive to bolster retrieval and reasoning over complex visual-textual data. The opportunity to contribute research at ETH through A/A* conference presentations and the Peers & Cheers forum, combined with knowledge-sharing through the ETH Analytics Club, an extension of my role as Machine Learning Lead of the Google Developer Student Club, is what makes ETH the ideal environment for my growth.

My short-term goal is to join a research lab of a leading AI company, such as Google DeepMind, OpenAI, or Anthropic, as a Research Scientist and help build next-generation multimodal foundation models that advance cross-modal reasoning and understanding. My long-term vision, building on this experience, is to establish a research group that concentrates on multimodal AI for education, developing systems that make quality learning accessible regardless of distance or finances, thereby bringing full circle my journey from a village where education was a privilege to empowering the masses with AI. With strong research experience and practical expertise in AI, I am confident that my depth and breadth of experiences will be valuable to the research community at ETH.